

Chapter 4

Torsion

4.1 Question Bank UNIT - 4

Quiz Questions

1. The diameter of a shaft is increased from 30 mm to 60 mm, all other conditions remaining unchanged. How many times is its torque carrying capacity increased?
2. Angle of twist of a shaft of diameter 'd' is inversely proportional to(in terms of 'd')
3. Shear strain in shaft subjected to torsion is inversely proportional to
4. The ratio of Min shear stress to Max shear stress for a thin walled closed structure is
5. A solid shaft transmits a torque T. The allowable shearing stress is τ . What is the diameter of the shaft?
6. A hollow shaft of outer dia 40mm and inner dia of 20mm is replaced by a solid shaft to transmit the same torque at the same maximum stress. What should be the diameter of the solid shaft?
7. A circular shaft is subjected to the combined action of bending, twisting and direct axial loading. The maximum bending stress σ , maximum shearing force 3σ and a uniform axial stress σ (compressive) are produced. The maximum compressive normal stress produced in the shaft will be.
8. One-half length of 50 mm diameter steel rod is solid while the remaining half is hollow having a bore of 25 mm. The rod is subjected to equal and opposite torque at its ends. If the maximum shear stress in solid portion is τ or, the maximum shear stress in the hollow portion is:
9. Torsional rigidity is defined as
10. The ratio of torque carrying capacity of a solid shaft to that of a hollow shaft is given by.....

Main Questions

1. A solid circular rod AB of diameter D and length L is fixed at both ends. A torque T is applied at a section X such that $AX = L/4$ and $BX = 3L/4$. What is the maximum shear stress developed
2. In a torsion test, the specimen is a hollow shaft with 50 mm external and 30 mm internal diameter. An applied torque of 1.6 kN-m is found to produce an angular twist of 0.4° measured on a length of 0.2 m of the shaft. The Youngs modulus of elasticity obtained from a tensile test has been found to be 200 GPa. Find the values of (i) Modulus of rigidity. (ii) Poissons ratio.
3. A hollow steel rod 200 mm long is to be used as torsional spring. The ratio of inside to outside diameter is 1 : 2. The required stiffness of this spring is 100 N.m /degree. Determine the outside diameter of the rod. Value of G is 8×10^4 N/mm²
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5. Calculate the maximum intensity of shear stress induced and the and the angle of twist in degrees in a solid shaft of 100 mm diameter, 10m long, transmitting a power of 112.5KW at 150 rpm. Take $G=82$ KN/mm²
6. A solid shaft is to transmit 300 kW at 120 rpm. If the shear stress is not to exceed 100 MPa, Find the diameter of the shaft, What percent saving in weight would be obtained if this shaft were replaced by a hollow one whose internal diameter equals 0.6 of the external diameter, the length, material and maximum allowable shear stress being the same?
7. Derive the Complete Torsion equation. Determine the torsional stiffness of a hollow shaft of length L and having outside diameter equal to 1.5 times inside diameter d . The shear modulus of the material is G .
8. A hollow shaft of diameter ratio 3/8 required to transmit 600 kW at 110 rpm, the maximum torque being 20% greater than the mean. The shear stress is not to exceed 63 MPa and the twist in a length of 3 m not to exceed 1.4 degrees. Determine the diameter of the shaft. Assume modulus of rigidity for the shaft material as 84 GN/m².
9. A 5 cm diameter solid shaft is welded to a flat plate by 1 cm filled weld. What will be the maximum torque that the welded joint can sustain if the permissible shear stress in the weld material is not to exceed 8 kN/cm² ? Deduce the expression for the shear stress at the throat from the basic theory.
10. A hollow shaft and a solid shaft construction of the same material have the same length and the same outside radius. The inside radius of the hollow shaft is 0.6 times of the outside radius. Both the shafts are subjected to the same torque. (i) What is the ratio of maximum shear stress in the hollow shaft to that of solid shaft? (ii) What is the ratio of angle of twist in the hollow shaft to that of solid shaft?